PURECHECK: IoT-Based Steroid Detection in Milk Using Multi-Sensor Analysis

***Abstract*—The increasing use of hormonal steroids in dairy farming has raised serious concerns regarding food safety and public health. Consumption of steroid-contaminated milk can lead to hormonal imbalance, endocrine disruption, and long-term health issues. This paper presents PURECHECK, an IoT-based system designed to detect and monitor steroid levels in milk using a multi-sensor approach. The system integrates pH, temperature, turbidity, TDS, and Near-Infrared (NIR) sensors to analyze milk quality in real time. Sensor data is processed and calibrated to estimate concentration levels, which are then transmitted to a cloud platform for analysis and storage. The system employs threshold-based safety checks and generates alerts when abnormal values are detected. The proposed solution ensures continuous monitoring, improved accuracy, and real-time decision-making, making it suitable for FSSAI-related monitoring and quality assurance applications.**

I. Introduction

Milk is one of the most widely consumed nutritional products, but its quality is often compromised due to the use of hormonal steroids in dairy farming. These substances are sometimes used to increase milk production, but their residues can pose serious health risks.

Milk plays a vital role in daily nutrition and is consumed by people of all age groups. However, ensuring its quality has become increasingly challenging due to the use of chemical substances, including hormonal steroids, in dairy farming practices.

Steroids are often used to increase milk yield and improve animal productivity. However, their residues in milk can have harmful effects on human health, including hormonal imbalance, reproductive disorders, and long-term metabolic complications.

In recent years, there has been a growing demand for real-time and continuous monitoring systems that can ensure milk safety. Conventional laboratory-based methods, while accurate, are not suitable for field-level deployment due to their complexity, cost, and time requirements.

To address these challenges, IoT-based monitoring systems have emerged as a promising solution. By integrating sensors, wireless communication, and cloud platforms, such systems enable continuous monitoring and early detection of abnormalities.

The PURECHECK system is designed to provide an efficient and scalable solution for monitoring milk quality using a multi-sensor approach combined with real-time data analysis and alert mechanisms.

A. Detection Efficiency

- Traditional detection methods such as laboratory testing are accurate but time-consuming and unsuitable for real-time monitoring. This creates a need for an efficient and continuous monitoring system.
- The proposed PURECHECK system addresses this problem using an IoT-based multi-sensor approach. By combining data from multiple sensors, the system improves detection reliability and enables real-time monitoring and alert generation.

II. Literature Survey

Pugazhenthi et al. developed an IoT-based milk monitoring system [1]. Ingole et al. proposed a multi-sensor system using ESP32 [2]. Rajakumar et al. introduced an IoT-based milk analysis system [3]. Fluorescence-based steroid detection methods are discussed in [4]. Advanced LC-MS/MS based detection is presented in [5]. IoT and machine learning based food spoilage detection is explored in [6].

The comparison of these approaches is summarized in Table I.

III. Proposed Methodology

- The PURECHECK system consists of multiple sensors connected to an ESP32 controller. The sensors measure parameters such as pH, temperature, turbidity, and TDS.
- The collected data is processed and calibrated to estimate concentration levels. The system then compares the values with predefined safe limits.
- The ESP32 controller performs real-time data acquisition from multiple sensors and processes the data using calibration models. The system continuously monitors the incoming data and compares it with predefined safety thresholds to detect abnormal conditions.
- The integration of IoT enables seamless communication between the sensing unit and cloud platform, allowing users to access real-time data remotely through a web interface.

A. Indirect Detection Explanation

- The proposed system performs indirect steroid detection by analyzing multiple physicochemical parameters of milk.
- Variations in pH, turbidity, and dissolved solids are correlated with abnormal chemical presence, enabling early indication of potential steroid contamination.

TABLE I: Literature Survey Comparison

Paper	Approach	Limitations	Improvement in PURECHECK
Pugazhenthi et al. (2023)	IoT-based milk monitoring using pH and temperature sensors	Limited to basic parameters and no detection of complex contaminants	Uses multi-sensor analysis including turbidity and TDS for better detection
Ingole et al. (2025)	Multi-sensor system with ESP32 for milk quality monitoring	Highly dependent on sensor accuracy and lacks advanced decision logic	Introduces calibration models and threshold-based automated decision making
Rajakumar et al. (2018)	IoT-based milk monitoring using multiple sensors and ESP32	Does not directly detect contaminants or pathogens	Improves detection using multi-sensor fusion and real-time monitoring
Varriale et al. (2015)	Fluorescence polarization assay for steroid hormone detection	Requires specialized equipment and has limited field applicability	Provides indirect detection suitable for real-time IoT deployment
He et al. (2020)	LC-MS/MS based detection of multiple steroid hormones	Expensive and requires skilled operation	Offers cost-effective and portable monitoring for field-level use
Mogilipalem et al. (2024)	IoT and machine learning based spoilage detection	Depends on ML model accuracy and training data quality	Uses threshold-based real-time detection without complex ML dependency
PURECHECK	IoT-based multi-sensor monitoring with real-time alert system	Indirect detection and not direct chemical identification	Improved reliability using multi-parameter analysis and real-time alerts

The complete hardware setup and sensor integration of the proposed system are illustrated in Fig. 2.

The system integrates pH, temperature, TDS, and turbidity sensors with an ESP32 microcontroller, enabling real-time data acquisition and cloud-based monitoring through WiFi.

IV. System Components

The PURECHECK system consists of multiple hardware and software components that work together to ensure effective monitoring of milk quality.

A. Sensor Module

The sensor module includes pH, temperature, turbidity, and TDS sensors. These sensors measure key physicochemical parameters of milk that are essential for detecting abnormalities.

B. Microcontroller Unit

An ESP32 microcontroller is used as the central processing unit. It collects data from all sensors, performs initial processing, and transmits data to the cloud.

C. Communication Module

The system uses Wi-Fi communication to transmit data to a cloud platform. This enables remote monitoring and real-time data access.

D. Alert System

The system includes LED indicators and a buzzer to provide immediate alerts. A green LED indicates safe conditions, while a red LED and buzzer indicate warning conditions.

E. Cloud and User Interface

The processed data is stored in a cloud database and can be accessed through a web dashboard. This allows users to monitor milk quality in real time.

V. Implementation Details

- The PURECHECK system is implemented using an ESP32 microcontroller integrated with multiple sensors including pH, temperature (DS18B20), turbidity, and TDS sensors. Each sensor is connected to the controller through appropriate analog and digital interfaces.
- The pH sensor provides analog voltage output corresponding to acidity levels, which is converted into digital values using the ADC of the ESP32. The DS18B20 temperature sensor communicates using a one-wire digital protocol, ensuring accurate temperature readings.
- The turbidity sensor measures the cloudiness of milk by detecting light scattering, while the TDS sensor measures dissolved solids present in the sample. These sensors collectively provide comprehensive data about the physicochemical properties of milk.
- The ESP32 processes the collected data and applies calibration algorithms to estimate concentration values. The processed data is transmitted to a cloud platform using Wi-Fi connectivity.
- A web-based dashboard is used for visualization, where users can monitor real-time data, view historical trends, and receive alerts. The alert system includes LED indicators and a buzzer for immediate notification of unsafe conditions.
- The implementation ensures low power consumption, cost efficiency, and scalability for real-world deployment.

VI. System Architecture

- The system architecture consists of multiple interconnected layers including sensing, processing, communication, and visualization.
- The sensing layer collects real-time data from multiple sensors. The processing layer, which includes the ESP32

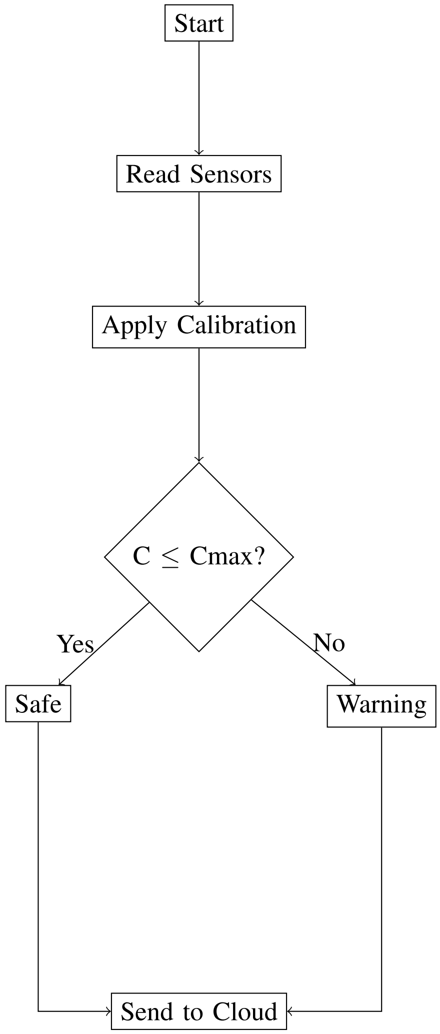

Fig. 1: Flowchart of PURECHECK System

controller, performs calibration and analysis of sensor data.

- The communication layer transmits processed data to the cloud using Wi-Fi connectivity. The cloud layer stores data and enables further analysis.
- Finally, the user interface layer provides real-time monitoring through a web dashboard, allowing users to visualize system performance and receive alerts.

VII. Mathematical Formulation

The proposed PURECHECK system uses calibration, threshold comparison, and alert logic to estimate steroid concentration in milk and determine safety status.

A. Sensor Calibration Formula

The raw sensor output is converted into standardized concentration units using a linear calibration model:

$$C = aS + b \tag{1}$$

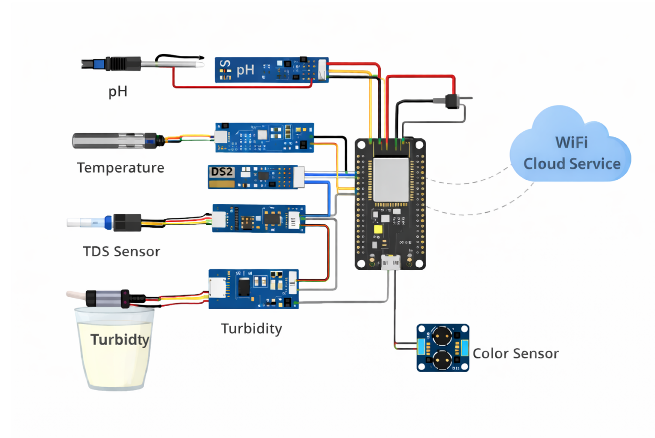

Fig. 2: Hardware architecture of the proposed PURECHECK system.

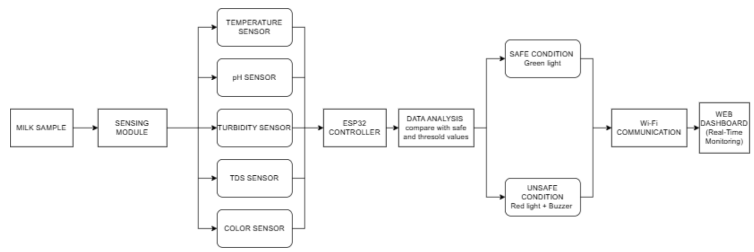

Fig. 3: Proposed PURECHECK System Architecture

where C is the steroid concentration in mg/L, S is the raw sensor output, a is the calibration coefficient, and b is the offset constant.

B. Safe Dosage Calculation

The safe allowable steroid quantity is calculated based on the milk quantity and permissible limit:

$$D_{\text{safe}} = Q \times L \tag{2}$$

where D_{safe} is the safe allowable quantity in mg, Q is the milk quantity in L, and L is the permissible steroid limit in mg/L.

C. Safety Check Condition

The detected concentration is compared with the maximum allowable limit to determine the milk safety status:

$$\text{Status} = \begin{cases} \text{Safe}, & C \leq C_{\max} \\ \text{Warning}, & C > C_{\max} \end{cases} \tag{3}$$

where C is the detected concentration and $C_{\max}$ is the maximum permissible concentration.

D. Percentage Exceedance

The percentage exceedance above the permissible limit is calculated as:

$$P = \left(\frac{C - C_{\max}}{C_{\max}} \right) \times 100 \tag{4}$$

where P is the percentage exceedance.

E. Decision Rule for Alert Generation

The alert decision is defined as:

$$\text{Alert} = \begin{cases} 1, & C > C_{\max} \\ 0, & C \leq C_{\max} \end{cases} \tag{5}$$

where Alert $= 1$ indicates a warning condition and Alert $= 0$ indicates a normal condition.

VIII. Comparison with Existing Methods

The comparison between the proposed PURECHECK system and existing milk quality monitoring methods is presented in Table II.

IX. Algorithm

The working procedure of the PURECHECK system can be summarized as follows:

1) Initialize all sensors and communication modules.
2) Read raw sensor values such as pH, temperature, turbidity, and TDS.
3) Apply calibration formula to convert raw sensor values into concentration levels.
4) Compare the calculated concentration with the predefined permissible limit.
5) If the concentration is within the safe limit, classify the sample as Safe.
6) If the concentration exceeds the limit, classify the sample as Warning.
7) Activate alert system (LED indicators and buzzer) based on the status.
8) Transmit data to the cloud platform for storage and monitoring.
9) Display real-time results on the web dashboard.

The algorithm ensures continuous monitoring, fast decision-making, and reliable detection of abnormal conditions in milk samples.

X. Dataset Collection

- The dataset used in the proposed PURECHECK system was collected through controlled experimental observations using sensor-based measurements on milk samples. The samples were analyzed under different conditions to simulate both normal and abnormal milk quality scenarios.
- Parameters such as pH value, temperature, turbidity, TDS, and calibrated concentration values were recorded. The dataset includes samples within permissible limits as well as samples exceeding threshold values to evaluate the effectiveness of the detection system.
- Data collection was performed at regular intervals to support real-time monitoring and trend analysis. The collected dataset was used for calibration, validation of the system, and graphical visualization of concentration variations.
- The dataset plays a crucial role in ensuring the reliability of the system by enabling accurate comparison between safe and unsafe conditions.

XI. Sensor Comparison

The comparison of sensors used in the PURECHECK system is shown in Table III.

XII. Sensor Selection Rationale

The selection of sensors in the PURECHECK system is based on accuracy, cost-effectiveness, and suitability for real-time monitoring.

The pH sensor is used to detect acidity variations in milk, which may indicate contamination or chemical imbalance. The temperature sensor ensures stable monitoring conditions and helps compensate for temperature-dependent variations.

The TDS sensor measures dissolved solids, which increase in the presence of impurities or additives. The turbidity sensor detects suspended particles and changes in milk clarity.

Although these sensors do not directly detect steroids, their combined analysis provides indirect evidence of abnormal chemical presence. This multi-sensor approach improves detection reliability and supports real-time monitoring.

XIII. Results and Discussion

The proposed system was evaluated using simulated and sensor-based sample readings collected under different milk quality conditions. The observed values indicate that variations in pH, temperature, and dissolved solids influence the calibrated concentration values and corresponding safety classification.

The results show that lower pH values and increased dissolved solids are associated with higher concentration estimates. When the detected concentration exceeds the permissible limit, the system changes the status from Safe to Warning and triggers the alert mechanism. This demonstrates that the proposed multi-sensor and threshold-based approach can support real-time indirect detection of potential steroid contamination in milk.

In addition to the tabular results, graphical analysis was performed to understand the variation of concentration levels over time. The observed trend indicates that an increase in turbidity and dissolved solids leads to higher estimated concentration values.

A. Performance Analysis

- The system successfully classified the milk samples into safe and warning categories based on the predefined threshold values. The alert mechanism was triggered accurately when the concentration exceeded the permissible limit.
- The integration of multiple sensors improved the reliability of detection compared to single-parameter monitoring systems. The system demonstrated consistent performance under different simulated conditions, indicating its potential for real-time deployment.
- The proposed approach provides a cost-effective and scalable solution for milk quality monitoring in dairy industries and supply chains.

TABLE II: Comparison of PURECHECK with Existing Methods

Feature	Existing Methods	PURECHECK System
Detection Method	Laboratory-based testing	IoT-based real-time monitoring
Monitoring Type	Offline / periodic	Continuous real-time monitoring
Parameters Used	Limited (pH, temperature)	Multi-sensor (pH, turbidity, TDS, temperature)
Detection Capability	Basic contamination detection	Indirect steroid detection using multi-parameter analysis
Response Time	Slow	Fast (real-time alerts)
Decision Making	Manual analysis	Automated threshold-based decision
Alert System	Not available	LED indicators and buzzer alerts
Data Storage	Limited or manual	Cloud-based storage and monitoring
Scalability	Limited	Highly scalable using IoT architecture
Cost	High (lab equipment required)	Cost-effective and portable
User Accessibility	Restricted to labs	Remote access via web dashboard

TABLE III: Comparison of Sensors Used in PURECHECK

Sensor	Parameter	Range	Accuracy	Voltage
pH Sensor	pH	0–14	± 0.1	5V
DS18B20	Temperature	-55 to 125°C	± 0.5°C	3–5V
TDS Sensor	Dissolved Solids	0–1000 ppm	± 10 ppm	3.3–5V
Turbidity Sensor	Clarity	0–1000 NTU	$\pm 5\%$	5V

TABLE IV: Sample Experimental Results

pH	Temp (°C)	TDS (ppm)	Conc. (mg/L)	Status
6.7	28	120	10	Safe
6.2	30	180	25	Warning
5.8	32	250	40	Warning

XIV. Advantages of Proposed System

- The PURECHECK system offers several advantages over traditional milk quality monitoring methods. One of the key advantages is real-time monitoring, which allows continuous observation of milk quality without the need for laboratory testing.
- The use of multiple sensors improves detection reliability by analyzing different physicochemical parameters simultaneously. This multi-sensor approach reduces the chances of false detection compared to single-parameter systems.
- The system is cost-effective and suitable for field-level deployment, making it accessible for small-scale dairy farmers as well as large dairy industries.
- The integration of IoT technology enables remote monitoring through cloud platforms, allowing users to access data anytime and anywhere. This improves transparency and decision-making in the milk supply chain.
- Additionally, the alert system ensures immediate notification of unsafe conditions, helping prevent contaminated milk from reaching consumers.

XV. Limitations of the System

- Despite its advantages, the PURECHECK system has certain limitations. The system performs indirect detection of steroid contamination based on variations in physicochemical parameters, which may not always provide exact identification of specific steroid compounds.
- The accuracy of the system depends on sensor calibration and environmental conditions. Variations in temperature or sensor drift may affect measurement accuracy over time.
- The system currently relies on predefined threshold values, which may vary depending on regional standards and milk composition.
- Additionally, the use of simulated data limits the validation of the system in real-world dairy environments. Further testing with real samples is required to improve system reliability.

XVI. Applications

- The PURECHECK system can be used in various applications within the dairy industry. It can be deployed in dairy farms to monitor milk quality at the source and ensure safe production practices.
- Milk collection centers can use the system to verify the quality of incoming milk before processing and distribution. This helps maintain quality standards in the supply chain.
- The system can also be integrated into transportation and storage units to monitor milk quality during transit.
- In addition, the system can be used in research laboratories and food safety organizations for continuous monitoring and analysis of milk quality.

XVII. Future Scope

The proposed system can be further enhanced by integrating machine learning algorithms for predictive analysis. This would enable more accurate detection of contamination patterns.

Additionally, the system can be extended to support large-scale dairy monitoring and integrated with mobile applications for user-friendly access. The use of advanced sensors such as biosensors and NIR spectroscopy can further improve detection accuracy.

XVIII. Conclusion

This paper presented PURECHECK, an IoT-based system for monitoring milk quality and detecting potential steroid contamination using a multi-sensor approach. The system enables real-time monitoring, data analysis, and alert generation.

The novelty of this work lies in integrating multi-sensor data fusion with IoT-based real-time monitoring for indirect detection of steroid contamination in milk.

The system demonstrated reliable performance under different conditions and successfully classified samples into safe and warning categories.

Future work includes integrating machine learning techniques for improved prediction accuracy, expanding the system for large-scale deployment, and incorporating advanced sensors for more precise detection.

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